

The discriminant validity of the Leader Reward and Punishment Questionnaire (LRPQ) and satisfaction with supervision: A two-sample, factor analytic investigation

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The conceptualization of leaders as reinforcement mediators is a relatively recent development, which has generated substantial interest in leadership research. Accompanying this new approach has been the development and use of perceived leader reinforcement questionnaires; a rapidly growing literature has developed as a result. This article reports a factor analytic investigation of what is probably the most commonly used and reliable measure in this domain – the Leader Reward and Punishment Questionnaire (LRPQ: Podsakoff, Todor, Grover & Huber, 1984). Using samples of hospital employees ($N = 375$) and bankers ($N = 297$), with principal axes factor extraction and varimax rotation, the results show relatively good discriminant validity when the LRPQ is factored with the Minnesota Satisfaction Questionnaire's 'satisfaction with supervision' items (MSQ: Weiss, Dawis, England & Lofquist, 1967). A consistent pattern of weak to moderate cross-loadings does exist, however, suggesting that research which employs affective dependent variables may need to interpret cautiously some of the relationships obtained. Avenues for future research are briefly discussed.

Many approaches to the study of leadership have emerged over the past 70 years (Bass, 1981), but one which has been suggested as particularly fruitful is the reinforcement approach (cf. House, 1988). The central concept involves the treatment of leaders as

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reinforcement mediators and the conceptualization of leader behaviour in terms of the administration of rewards and punishments to subordinates contingent on their performance. The reinforcement perspective has generated considerable theoretical and empirical interest, and a substantial literature has developed as a result (Williams & Podsakoff, 1988).

Most recent investigations in this domain have employed the Leader Reward and Punishment Questionnaire (LRPQ) of Podsakoff, Todor & Skov (1982; reported in Podsakoff *et al.*, 1984). The LRPQ built on the earlier work of Sims & Szilagyi (1975) by deleting, adding and modifying items, so that both contingent (performance-related) and non-contingent leader reward and punishment behaviours are assessed. As currently constituted, the four dimensions assessed by the LRPQ are: Contingent Reward behaviour (CR: 10 items), Non-Contingent Reward behaviour (NCR: four items), Contingent Punishment (CP: five items), and Non-Contingent Punishment (NCP: four items) (see the Appendix for the items of the LRPQ).

Many analyses have been conducted which have demonstrated acceptable internal consistency, test-retest reliability, factor structure and both concurrent and predictive validity for these leader reinforcement scales (Podsakoff & Schriesheim, 1984 *a, b*; Podsakoff *et al.*, 1984). However, research exploring the broader domain of psychometric issues is clearly needed to increase confidence in these measures and their accumulated findings (cf. Schwab, 1980). One psychometric property of the LRPQ that should be investigated is its discriminant validity, i.e. whether the subdimensions (subscales) of the LRPQ are sufficiently differentiated from other, supposedly different, constructs (Nunnally, 1978; Schwab, 1980). Since the LRPQ is designed to obtain subordinates' descriptions of the perceived behaviour of their supervisors, it seems reasonable to ask whether these descriptions are sufficiently differentiated from affective reactions to those behaviours. Research in other domains has found serious confounding between highly correlated but supposedly distinct descriptive and affective measures, so the question of artifactual relationships due to poor discriminant validity naturally arises. The current research therefore followed the approach suggested by Ferratt, Dunham & Pierce (1981) and undertook a factor analytic exploration of the empirical distinctiveness (discriminant validity) of the four LRPQ subscales, and a commonly used and well developed measure of job satisfaction – the Minnesota Satisfaction Questionnaire (MSQ: Weiss *et al.*, 1967).

The MSQ is employed because it is intended to be a strictly evaluative or affective instrument, so that it should be empirically distinct from the LRPQ. In addition, satisfaction with supervision has been used by more respondents than any other dependent variable in leader reinforcement research, and the meta-analytically estimated 'true correlation' (corrected for artifacts) between CR and supervision satisfaction is .64 (Williams & Podsakoff, 1988). The MSQ also possesses excellent internal consistency reliability and other psychometric properties (Weiss *et al.*, 1967), and its proportion of true score (non-error and non-method) variance is higher than several other established satisfaction measures (Schriesheim, Tetrault, Kinicki & Carson, 1989).

Method

Sample and procedure

Two samples were employed for this study, to reduce the likelihood that obtained results are artifacts and to

enhance the generalizability of findings. The first sample consisted of 375 full-time employees of a large southern US hospital. The average age was 34 years, average organizational tenure was 4.5 years, and 73 per cent were female. A total of 91 per cent of the hospital respondents had graduated from high school and 40 per cent had college degrees or graduate training. The second sample consisted of 297 mid-level managers employed at over 100 US financial institutions. The average age of the respondents was 32.3 years and 69 per cent were male. The respondents' average length of tenure in the profession was 4.3 years, and over 90 per cent of the respondents were college graduates and all had at least some college education. Questionnaires were administered to the hospital employees in their usual work settings, during normal work hours, while the banking sample was administered the same measures at the beginning of a two-week residential industry-sponsored management development programme.

Measures

Leader reward and punishment behaviour. Perceived leader reward and punishment behaviour was assessed by the 23-item instrument developed by Podsakoff and Skov and presented in Podsakoff *et al.* (1984, p. 38). Following Podsakoff and his associates, a seven-point Likert scale, ranging from 'strongly disagree' to 'strongly agree', was employed in both samples here.

Satisfaction. Satisfaction with supervision was measured by the two five-item subscales, Supervisor-Human Relations (SSHR) and Supervisor-Technical Ability (SSTA), of the long (100-item) form of the MSQ (Weiss *et al.*, 1967); these items are also shown in the Appendix.

Analytical procedures

Coefficient alpha internal consistency reliabilities, means, standard deviations and intercorrelations were computed for the six scales in both samples (separately) and found to be well within the ranges typically reported in the literature (all coefficient alphas were in excess of .70). Thus, the samples employed in this investigation should not have produced atypical findings. Also, the LRPQ was factored by itself in both samples (separately), and showed excellent dimensional structure.

The main analyses of this study, however, involved factoring the LRPQ and MSQ together, to explore the discriminant validity of the LRPQ. For ease of interpretation and conformity to Podsakoff *et al.*'s (1984) procedures (and common convention), these analyses used iterated principal axes factoring, with squared multiple correlations as initial communality estimates and varimax rotations. For these analyses the LRPQ and satisfaction with supervision items were factored together in each sample (the two samples were not combined).*

Results

Hospital sample

Eigenvalue-one, scree (Cattell, 1966) and discontinuity (Harman, 1967) criteria, as well as the theoretical structure of the LRPQ and MSQ, all indicated that it was appropriate to extract five factors from the hospital data. Extracting these and subjecting them to a varimax rotation produced the results shown in the columns headed 'A' in Table 1.

As seen in Table 1, the factor structure of the LRPQ is generally good and shows considerable distinctiveness between its subdimensions. Using the .40 criterion level, which appears to be most commonly used in judging factor loadings as meaningful (Ford, MacCallum & Tait, 1986), it seems clear that factor 1 measures satisfaction with

* It should be mentioned that a second set of analyses were also conducted on these data, employing LISREL maximum likelihood confirmatory factor analysis (Joreskog & Sorbom, 1984) to assess statistically the goodness of fit of the LRPQ items to their theoretical *a priori* dimensions. To conserve space, these analyses are not reported here but they very strongly confirmed the theoretical dimensionality of the LRPQ in both samples (full tabular results are available from the authors on request).

Table 1. Five-factor varimax solution for LRPQ and MSQ supervision items^a

Scale and item no.	1 ^b		2		3		4		5		h ²	
	A	B	A	B	A	B	A	B	A	B	A	B
CP 14	15	09	11	11	-19	19	34	<u>66</u>	-19	-08	22	49
CR 05	30	13	<u>65</u>	<u>72</u>	25	-24	15	11	15	01	62	61
NCR 21	14	-06	28	-04	-13	04	02	-05	<u>47</u>	<u>78</u>	34	62
CR 09	-24	22	-22	<u>71</u>	-37	-18	02	13	01	-04	25	61
CR 07	34	32	<u>51</u>	<u>53</u>	08	-02	11	16	23	11	45	42
CP 13	05	14	14	03	-10	-05	<u>60</u>	<u>81</u>	-05	-10	39	70
CR 08	<u>43</u>	<u>50</u>	<u>56</u>	<u>57</u>	21	-05	14	04	14	-04	59	59
NCP 18	-16	-24	-13	-28	<u>-73</u>	<u>67</u>	02	11	02	-09	58	61
CR 03	37	34	<u>61</u>	<u>65</u>	11	-03	38	08	05	01	67	55
NCR 22	10	02	05	-02	07	-13	01	-31	<u>67</u>	<u>59</u>	47	46
CR 02	38	21	<u>63</u>	<u>73</u>	21	-18	16	12	29	01	70	63
CR 04	36	28	<u>71</u>	<u>74</u>	19	-01	29	09	25	-07	82	65
NCR 20	12	-07	17	01	-12	04	02	-11	<u>65</u>	<u>82</u>	48	70
NCP 19	-23	-08	-06	-22	<u>-51</u>	<u>77</u>	07	-03	22	05	38	65
CP 11	11	09	16	15	-01	-12	<u>63</u>	<u>78</u>	10	-05	44	65
CR 01	36	27	<u>65</u>	<u>81</u>	24	-05	31	11	18	-01	74	74
CP 12	07	01	-01	08	01	11	<u>78</u>	<u>67</u>	09	-16	62	49
NCR 23	10	-04	17	38	-14	13	-10	-20	<u>53</u>	<u>44</u>	35	40
CR 10	-31	25	-37	<u>68</u>	<u>-43</u>	-25	18	05	-04	00	45	59
CR 06	30	33	<u>59</u>	<u>65</u>	11	-11	16	13	18	-09	50	57
CP 15	13	06	20	17	-07	-03	<u>68</u>	<u>77</u>	-08	-08	53	64
NCP 16	-26	<u>-41</u>	-15	-07	<u>-60</u>	<u>42</u>	08	10	03	05	45	36
NCP 17	-34	-25	-14	-22	<u>-67</u>	<u>48</u>	08	06	13	<u>45</u>	60	55
SSHR 10	<u>58</u>	<u>60</u>	38	<u>54</u>	26	-16	08	02	20	06	60	68
SSTA 15	<u>65</u>	<u>71</u>	39	04	15	-21	17	12	10	01	64	57
SSHR 30	<u>73</u>	<u>73</u>	26	37	22	07	09	-05	10	00	66	67
SSTA 35	<u>72</u>	<u>79</u>	39	22	21	-06	04	07	07	-09	73	68
SSHR 50	<u>69</u>	<u>68</u>	22	30	20	-18	09	10	10	-08	58	60
SSTA 55	<u>63</u>	<u>64</u>	25	21	24	-19	13	16	21	-10	58	52
SSHR 70	<u>79</u>	<u>68</u>	19	29	20	-07	11	02	09	-03	71	55
SSTA 75	<u>77</u>	<u>76</u>	26	30	16	-14	14	08	03	-08	71	69
SSHR 90	<u>75</u>	<u>76</u>	16	24	23	-02	08	11	16	-01	67	64
SSTA 95	<u>71</u>	<u>72</u>	27	35	19	07	19	09	07	-04	65	65
Unrotated eigen-value	12.3	11.1	3.2	3.2	2.4	1.8	1.2	1.8	1.1	2.3	20.2	19.6
% Total variance	37.3	33.6	9.8	9.8	7.2	5.3	3.6	5.3	3.3	6.8	61.2	59.2

^a The items are presented in the same order as on the survey questionnaire but are numbered in accordance with the original sources (Podsakoff *et al.*, 1984; Weiss *et al.*, 1967); the LRPQ and MSQ items are shown in the Appendix, also numbered according to the original sources.

^b The columns headed 'A' present the hospital sample loadings, while those headed 'B' give the loadings for the bankers. Decimal points are omitted from the factor loadings. Loadings of .40 magnitude or greater are underlined.

supervision, while factor 2 assesses CR. Factor 3 is concerned with NCP, while factor 4 measures CP; factor 5 assesses NCR.

Looking at specific items, it can be seen that CP 14 has only a .34 loading on its appropriate factor (although it has no cross-loading greater than $-.19$ in magnitude). Also, items CR 09 and 10 do not load at the .40 level on their appropriate factor (factor 2) and both have cross-loadings on the NCP factor (factor 3) which are larger than their appropriate loadings.

Nine of the 10 CR items (all but CR 09) have weak to moderate cross-loadings on the satisfaction factor (factor 1), as does item NCP 17. However, only the loading for CR item 08 is in excess of .40 and the appropriate loadings are substantially higher than the cross-loadings for all but CR 08 and 10. Although the factor loadings of the satisfaction with supervision items are generally good, three (SSHR 10 and SSTA 15 and 35) also have weak cross-loadings on the CR factor (factor 2).

Banker sample

As with the hospital sample, eigenvalue, scree and discontinuity criteria, as well as *a priori* dimensionality, all suggested the extraction of five factors for the bankers. Five factors were therefore extracted, a varimax rotation employed, and the factors then reordered to correspond to those obtained in the hospital sample (based on the similarity of the loadings on the items); the results are shown in the columns headed 'B' in Table 1.

As shown in Table 1, the pattern of factor loadings for the bankers provides reasonable support for the *a priori* dimensionality of the LRPQ. Only NCP 17 has a moderate cross-loading on another LRPQ factor, although NCR 22 and 23 also have weak cross-loadings on other LRPQ factors.

Looking at the CR and satisfaction items, however, discloses a clear pattern of cross-loadings for both. Here, three of the 10 CR items (CR 03, 06 and 07) can be seen to have weak cross-loadings on satisfaction (factor 1), while CR 08 and NCP 16 have moderate cross-loadings (these two items also do not have appropriate loadings which are appreciably higher than these cross-loadings). Although the loadings of the satisfaction with supervision items are generally good, SSHR 30 and 50 and SSTA 75 and 95 have weak cross-loadings on CR (factor 2) and SSHR 10 has a reasonably high (.54) cross-loading.

Comparing the results across the two samples indicates an excellent factor structure for the LRPQ in general. However, weak and moderate cross-loadings occur between the CR and satisfaction items in both samples. In particular, CR 03, 06 and 07 and SSHR 10 have consistent weak cross-loadings and CR 08 has moderate cross-loadings in both samples.

Discussion

The results pertaining to the discriminant validity of the LRPQ must be viewed as supporting the LRPQ but also as indicating the need for caution when its CR scale is used in studies which involve satisfaction with supervision as measured by the MSQ or similar supposedly affective satisfaction measures.

Deleting CR items 03, 06, 07 and 08 and rerunning the analyses of the current study resulted in improvements in the factor structures of both samples; virtually all cross-loadings of LRPQ items on satisfaction disappeared. The coefficient alpha reliability of the

CR scale also stayed above .70 in both samples, apparently because the remaining six CR items are a reasonably homogeneous sample from the domain of contingent leader reward behaviour. Thus, while it would be premature to suggest that CR items 03, 06, 07 and 08 be deleted from the LRPQ to improve its discriminant validity, it does not seem unreasonable to suggest that further research is needed on this issue, so that the evidence which accumulates on leader reward behaviour is not tainted by suspicions of artifactual relationships. This seems particularly important given the fact that some supervision satisfaction measures, such as the Job Descriptive Index (Smith, Kendall & Hulin, 1969), have more intentionally descriptive item content than does the MSQ. This, of course, might heighten item cross-loading patterns and aggravate concerns about artifactual relationships.

In conclusion, the current investigation suggests that the confounding of descriptions (reported cognitions) and affect may be somewhat of a problem in leader reinforcement research. It may be that the wording of survey questionnaires creates or contributes to cognition-attitude confounding, that survey respondents cannot or do not distinguish between what they believe and what they feel, or that some combination of these factors is at work. In any event, research exploring the sources of such confounding may be useful and help us better understand basic organizational processes.

As a final comment, it is appropriate to highlight that the current research should also be viewed as supporting the discriminant validity of the LRPQ on an overall basis, as well as its *a priori* theoretical structure. Too often measures are used in organizational research without adequate psychometric investigation (Schwab, 1980), and too often psychometric examination leads to abandonment rather than development and improvement. Thus, while further investigation of the psychometric properties of the LRPQ seems desirable, it should not be abandoned as a useful instrument for studying leaders as reinforcement mediators. On the contrary, given the good psychometric properties demonstrated for the LRPQ, research aimed at improving it and subjecting it to further scrutiny clearly seems warranted.

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Appendix

*Leader Reward and Punishment Questionnaire Items^a**Contingent Reward behaviour (CR)*

- CR 01 My supervisor always gives me positive feedback when I perform well
- CR 02 My supervisor gives me special recognition when my performance is especially good
- CR 03 My supervisor would quickly acknowledge an improvement in the quality of my work
- CR 04 My supervisor commends me when I do a better than average job
- CR 05 My supervisor personally pays me a compliment when I do outstanding work
- CR 06 My supervisor informs his/her boss and others when I do outstanding work
- CR 07 If I do well, I know my supervisor will reward me
- CR 08 My supervisor would do all that he/she could to help me go as far as I would like to go in this organization if my work is consistently above average
- CR 09 My good performance often goes unacknowledged by my supervisor (reverse-scored)
- CR 10 I often perform well in my job and still receive no praise from my supervisor (reverse-scored)

Contingent Punishment behaviour (CP)

- CP 11 If I performed at a level below that which I was capable of, my supervisor would indicate his/her disapproval
- CP 12 My supervisor shows his/her displeasure when my work is below acceptable levels
- CP 13 My supervisor lets me know about it when I perform poorly
- CP 14 My supervisor would reprimand me if my work was below standard
- CP 15 When my work is not up to par, my supervisor points it out to me

Non-Contingent Punishment behaviour (NCP)

- NCP 16 My supervisor frequently holds me accountable for things I have no control over
- NCP 17 My supervisor is often displeased with my work for no apparent reason
- NCP 18 My supervisor is often critical of my work, even when I perform well
- NCP 19 I frequently am reprimanded by my supervisor without knowing why

Non-Contingent Reward behaviour (NCR)

- NCR 20 Even when I perform poorly, my supervisor often commends me
- NCR 21 My supervisor is just as likely to praise me when I do poorly as when I do well
- NCR 22 Even when I perform poorly on my job, my supervisor rarely gets upset with me
- NCR 23 My supervisor frequently praises me even when I don't deserve it

*Minnesota Satisfaction Questionnaire items^b**Satisfaction with Supervisor Human Relations (SSHR)*

- SSHR 10 The way my supervisor and I understand each other
- SSHR 30 The way my boss handles his/her subordinates
- SSHR 50 The way my boss backs his/her subordinates up (with top management)
- SSHR 70 The way my boss takes care of complaints brought to him/her by his/her subordinates
- SSHR 90 The personal relationship between my boss and his/her subordinates

Satisfaction with Supervisor Technical Ability (SSTA)

- SSTA 15 The technical 'know-how' of my supervisor
- SSTA 35 The competence of my supervisor in making decisions
- SSTA 55 The way my boss delegates work to others
- SSTA 75 The way my boss provides help on hard problems
- SSTA 95 The way my boss trains his/her subordinates

^a From Podsakoff *et al.* (1984, p. 38).^b From Weiss *et al.* (1967, pp. 32-35).

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